

Amine Maazizi

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Website | GitHub | LinkedIn

Research Interests

Representation learning, multimodal learning, and geometric deep learning, with interests in foundation models and structure-aware machine learning.

Education

ENS Paris-Saclay <i>Master in Mathematics, Vision and Learning (MVA)</i> Relevant coursework: Probabilistic Graphical Models, Deep Generative Models, Geometric Data Analysis, Representation Learning, Deep Learning for Medical Imaging, Graphs in Machine Learning, Multimodal Explainable AI.	2025 – 2026
ENSTA Paris, Institut Polytechnique de Paris <i>Engineering Degree in Computer Science and Artificial Intelligence</i> Relevant coursework: Machine Learning, Statistical Learning, Deep Learning, Computer Vision, Language Models, Reinforcement Learning.	2023 – 2026 GPA: 4.05/4.3 , Top 5%
CPGE Moulay Youssef, Rabat, Morocco <i>Preparatory Classes (MPSI / MP*)</i> Ranked among the top 5% in national competitive entrance examinations.	2021 – 2023 Top 5% nationally

Research Experience

EPFL, Artificial Intelligence in Molecular Medicine Lab, Lausanne, Switzerland <i>Incoming Research Intern</i> Supervisors: Dr. Lukas Klein Investigating foundation models for cellular perturbation modeling through the VirTues-Sim framework, framing single-cell response prediction as a generative world model over cellular states. Developing methods to learn consistent latent dynamics under interventions, modeling tissue-level evolution as structured state transitions, and evaluating generalization across heterogeneous biological contexts.	Apr 2026 – Sep 2026
Institut Pasteur, Biomedical Image Analysis Unit, Paris, France <i>Research Intern and Collaborator</i> Supervisor: Prof. Giacomo Nardi Built a geometry-aware shape-matching framework for biological membranes using Helfrich energy, reducing mesh-projection error relative to linear-elasticity baselines and supporting a curvature-driven interpretation of EHT.	Jun 2025 – Dec 2025
NAIST, Cybernetics and Reality Engineering Lab, Remote <i>Research Collaborator</i> Supervisor: Prof. Hideaki Uchiyama Authored a structured survey of text- and image-conditioned 3D generation by synthesizing reconstruction paradigms across the literature, showing the field's shift from geometric inductive bias toward 2D generative priors.	Oct 2024 – Dec 2024

Selected Publications and Research Outputs

not-MIWAE: Deep Generative Modelling with Missing Not at Random Data and a Supervised MNAR Extension A. Maazizi, A. Gasse, E. Melzani <i>Preprint, under review</i>	Manuscript
Morphodynamic Study of Hematopoietic Stem Cell Emergence Using Shape Matching with Mechanical Constraints G. Nardi, A. Maazizi, et al. <i>Manuscript in preparation</i>	
From Geometric Inductive Bias to 2D Generative Priors: A Structured Survey of Deep Learning for 3D Reconstruction A. Maazizi	

Technical report, NAIST CaRE Lab	Link
Efficient Wasserstein Barycenter Computation with Application to Texture Mixing A. Maazizi, E. Melzani, S. Oukaci Technical report, MVA Geometric Data Analysis	Link
Clairvoyance: Smart Glasses for the Visually Impaired in Indoor Environments G. Dupuis, A. Maazizi, et al. Technical report, ENSTA Paris	Link

Software and Projects

PointLLM Contrastive Alignment	GitHub
Investigated contrastive versus generative objectives for point cloud-language alignment in a PointLLM architecture, found improved cosine similarity and tighter t-SNE embedding alignment with the contrastive objective.	
Geometry-Aware Shape Matching	Private Code
Built, in collaboration with non-technical biologists at Institut Pasteur, a geometry-aware mesh-matching framework with configurable mechanical priors, enabling constrained shape matching on biological microscopy data.	
not-MIWAE Pytorch	GitHub / PyPI
Built a PyTorch package to unify MIWAE, not-MIWAE, supMIWAE, and sup-not-MIWAE in a single missing-data framework, making MNAR modeling easier to reproduce and extend.	
Self-Supervised Nuclei Segmentation	GitHub
Built a self-supervised nuclei segmentation pipeline by pretraining an encoder with BYOL and transferring representations to a U-Net model, evaluating improvements in Dice/IoU performance and training stability relative to supervised baselines.	
BiDA MRI Super-Resolution	GitHub
Reproduced a simplified version of BiDA for rodent brain MRI super-resolution by training DDPM priors on CAMRI rat data and evaluating on an OOD mouse dataset, achieving a +2.6 dB PSNR improvement over cubic interpolation.	

Awards, Honors and Talks

Poster, <i>not-MIWAE and a Supervised MNAR Extension</i> , ENS Paris-Saclay Atrium, Poster	Dec 2025
Finalist, Best Research Project Award, ENSTA Paris	2025
Moroccan Excellence Scholarship, Ministry of Higher Education	2023

Technical Skills

Programming: Python, C, C++
Machine Learning: PyTorch, JAX, TensorFlow, PyTorch Geometric
Scientific Computing: NumPy, SciPy
Infrastructure: Linux, Git, HPC (SLURM)

Languages

Arabic (Native) | French (Bilingual) | English (Fluent) | Spanish (Elementary)